

LOG ANALYSIS BASICS

Mastering Log Analysis for Cybersecurity: A Hands-On Journey from Zero to Hero

LOG ANALYSIS WITH WINDOWS EVENT VIEWER AND LINUX [/VAR/LOG – AUTH.LOG]

hands-on learning journey into log analysis, an essential SOC skill. This guide covers detection of suspicious activities across Windows and Linux systems, backed by real-world labs and key lessons.

Day 1: Introduction to Log Analysis

Day 2: Detecting Failed Logins (Windows Event ID 4625)

Day 3: Detecting Suspicious PowerShell Commands (Event ID 4103)

Day 4: Detecting Port Scans (UFW Logs)

Day 5: Detecting SSH Brute-Force Attacks (auth.log)

Day 1: Introduction to Log Analysis

Objective:

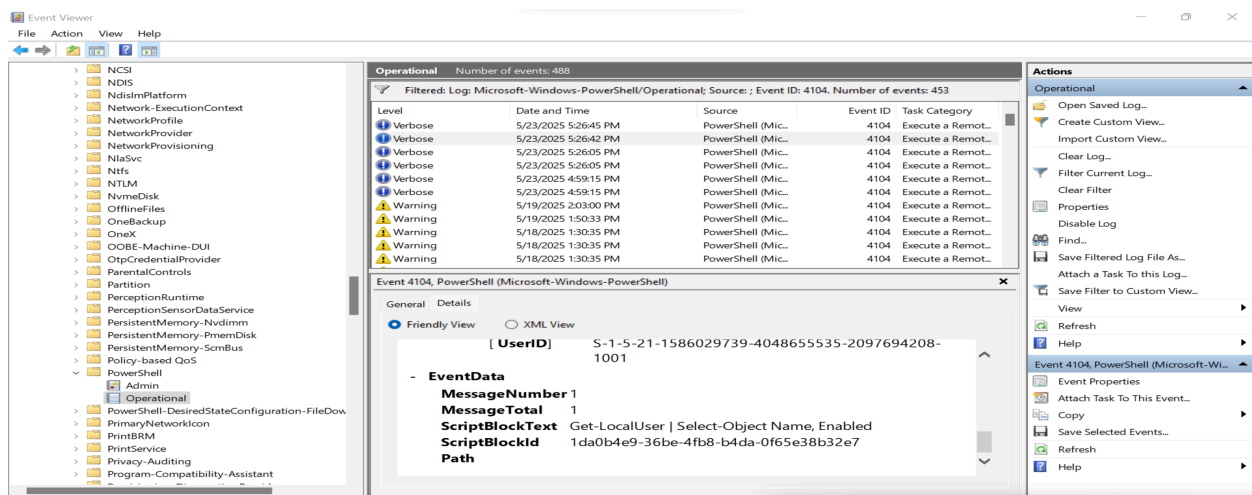
Understand logs as the backbone of cybersecurity operations—capturing system events, authentication attempts, and suspicious activities.

What I Did:

Explored key log sources:

Windows Event Viewer

Simulated a suspicious PowerShell command and detected it in Windows Event Viewer (Event ID 4104).



Key Takeaways:

1. Logs record events with timestamps, sources, and actions.
2. Event IDs are essential markers for specific activities.
3. Logs are critical for detecting incidents, performing investigations, and ensuring compliance.

Day 2: Detecting Failed Logins (Event ID 4625)

Objective:

Learn to detect unauthorized access attempts through Windows Security Logs.

What I Did:

Simulated a failed login attempt:

```
net use \\127.0.0.1\IPC$ /user:haxuser WrongPassword
```

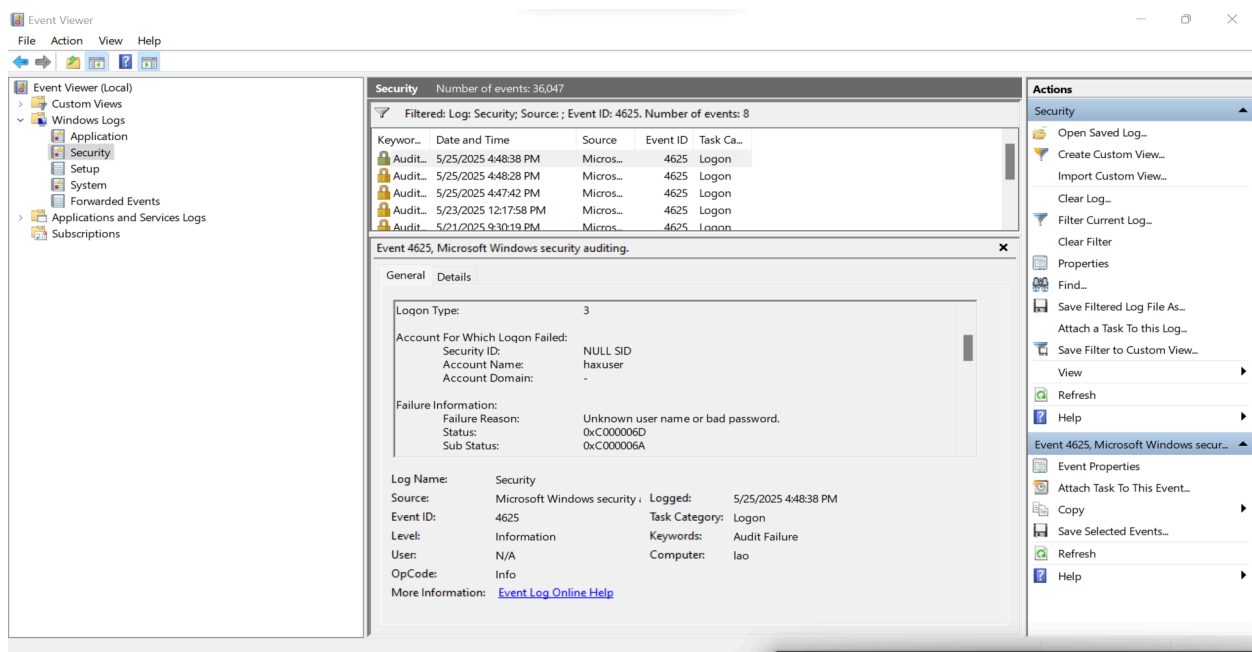
Filtered Security Logs by Event ID 4625 and identified:

User Name: haxuser

Logon Type: 3 (Network)

Source IP: 127.0.0.1

Reason: Bad username or password



```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\LAOTSE> net use \\127.0.0.1\IPC$ /user:haxuser WrongPassword
System error 1326 has occurred.

The user name or password is incorrect.

PS C:\Users\LAOTSE> net use \\127.0.0.1\IPC$ /user:haxuser WrongPassword
System error 1326 has occurred.

The user name or password is incorrect.

PS C:\Users\LAOTSE> net use \\127.0.0.1\IPC$ /user:haxuser WrongPassword
System error 1326 has occurred.

The user name or password is incorrect.

PS C:\Users\LAOTSE>
```

Key Takeaways:

1. Event ID 4625 highlights failed login attempts—a key indicator of brute-force attacks.
2. Understanding logon types helps determine attack methods (e.g., network vs. local).

Day 3: Detecting Suspicious PowerShell Commands (Event ID 4103)

Objective:

Detect malicious command execution using PowerShell logging.

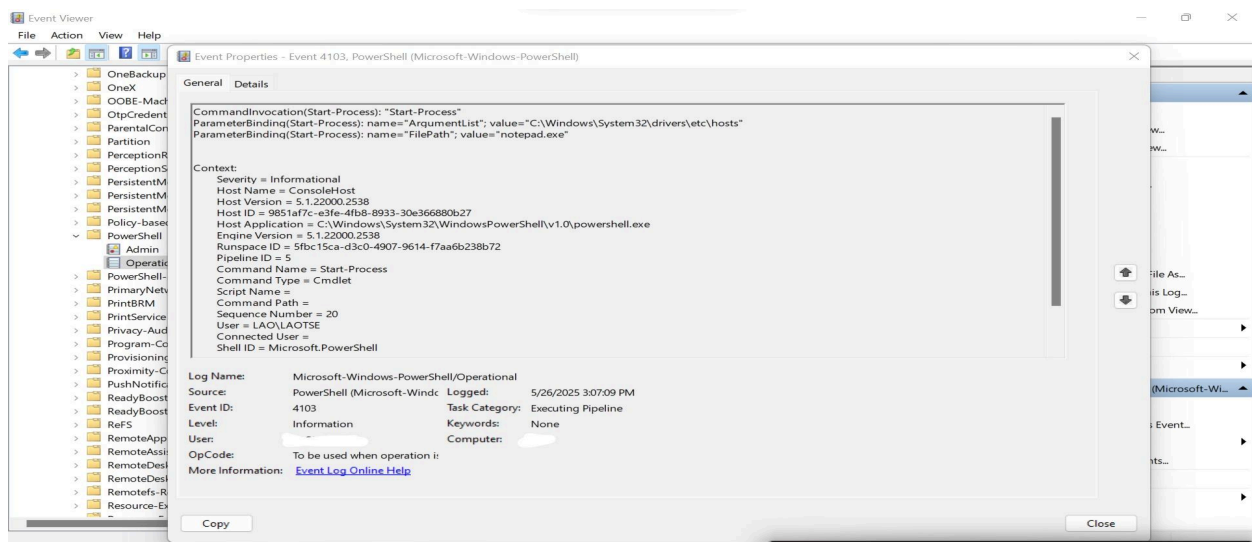
What I Did:

Enabled PowerShell Module Logging and Script Block Logging.

Simulated a command:

```
Start-Process "notepad.exe" -ArgumentList  
"C:\Windows\System32\drivers\etc\hosts"
```

Detected in Event Viewer (Applications and Services Logs > PowerShell > Operational) with Event ID 4103.



Example of LOLBAS Tools These are legitimate Windows tools that attackers often abuse for stealthy malicious actions.

🔧 Tool	📁 Path	🚩 Abuse Technique
powershell.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	Execute payloads, download malware, bypass AV
certutil.exe	C:\Windows\System32\certutil.exe	Download files using: <code>certutil -urlcache -f</code>
mshta.exe	C:\Windows\System32\mshta.exe	Execute malicious HTML apps or remote scripts
regsvr32.exe	C:\Windows\System32\regsvr32.exe	Load and execute remote/local DLLs
rundll32.exe	C:\Windows\System32\rundll32.exe	Execute DLLs or scripts to evade detection
wmic.exe	C:\Windows\System32\wbem\wmic.exe	Execute commands, gather system info
bitsadmin.exe	C:\Windows\System32\bitsadmin.exe	Download/upload files silently
msbuild.exe	C:\Windows\Microsoft.NET\Framework\v4.0.30319\msbuild.exe	Execute malicious C# code in project files
installutil.exe	C:\Windows\Microsoft.NET\Framework\v4.0.30319\installutil.exe	Run code during .NET assembly install
schtasks.exe	C:\Windows\System32\schtasks.exe	Create scheduled tasks for persistence

Key Takeaways:

1. PowerShell logs reveal post-exploitation activity.
2. Monitor for LOLBAS usage—legitimate tools used for malicious purposes.

Day 4: Detecting Port Scans (UFW Logs)

Objective:

Simulate and detect network reconnaissance using Linux UFW logs.

What I Did:

Ran a port scan from Kali:

```
nmap -p80 TARGET-IP
```

Configured UFW on the target machine:

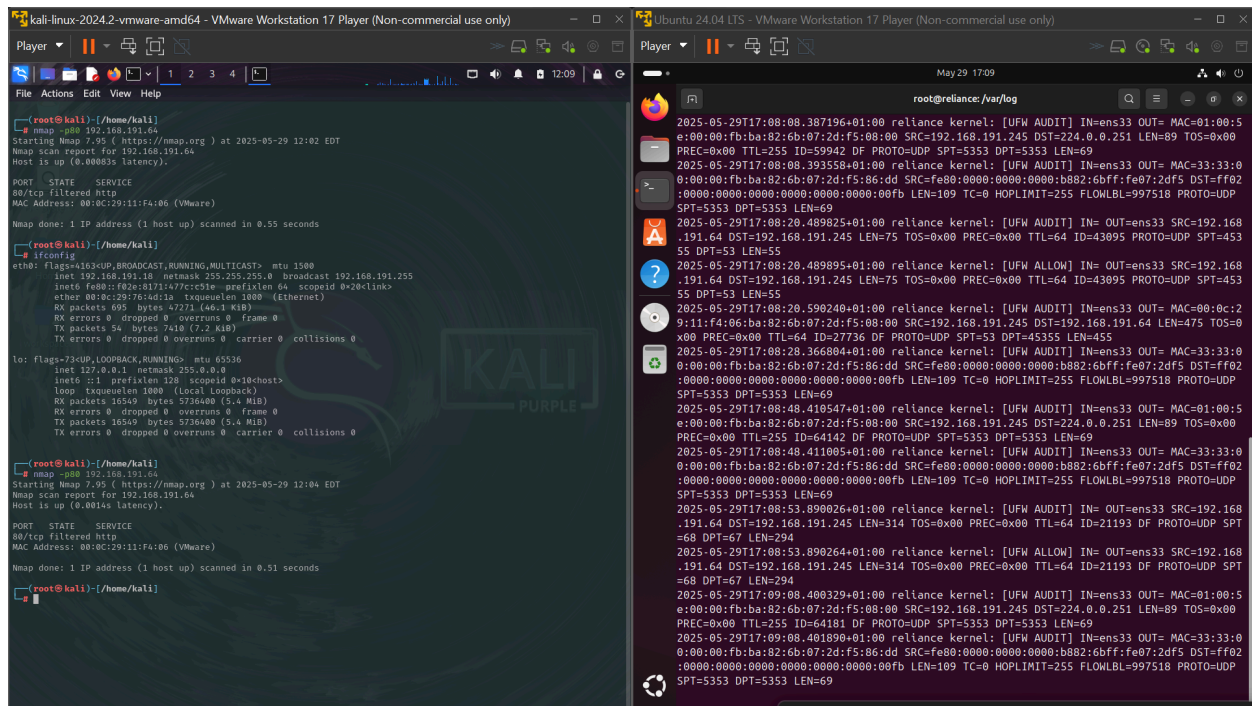
```
sudo ufw enable
```

```
sudo ufw logging on
```

```
sudo ufw deny from <attacker-ip> to any port 80
```

Detected scan attempts in:

```
/var/log/ufw.log
```



Key Takeaways:

1. Port scans are early signs of potential attacks.
2. UFW logs provide visibility into incoming connections and blocked attempts.

Day 5: Detecting SSH Brute-Force Attacks (auth.log)

Objective:

Identify brute-force SSH attacks on Linux systems.

What I Did:

Simulated an SSH brute-force attack using Hydra.

Monitored /var/log/auth.log for:

Multiple failed login attempts from a single IP.

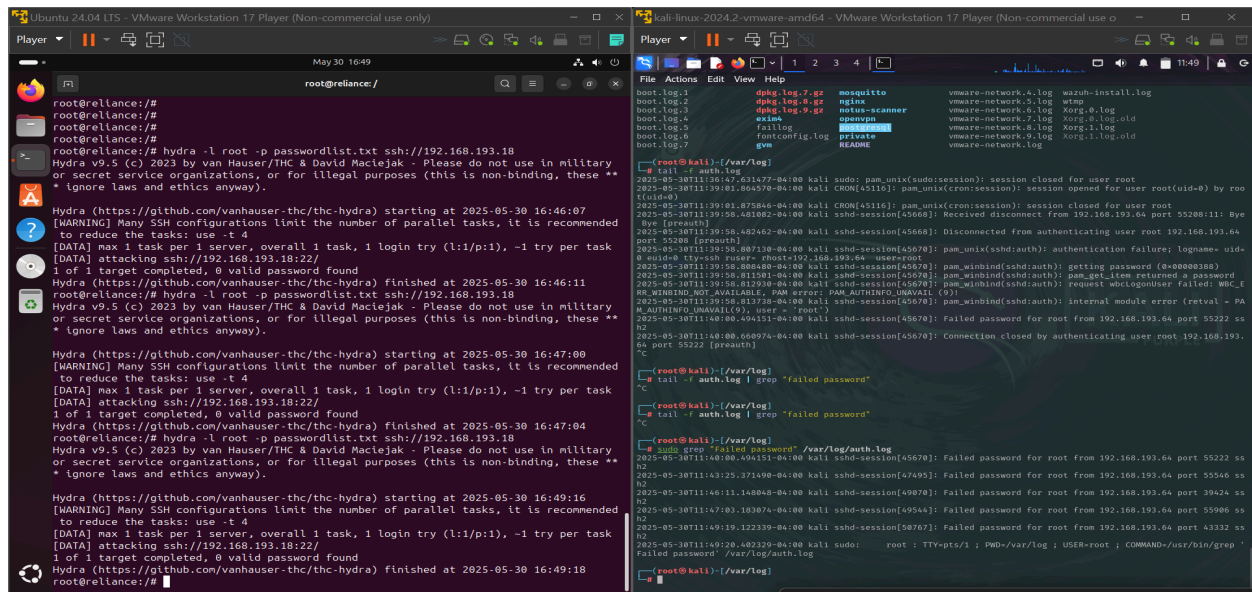
Repeated attempts on sensitive users like root.

Challenges Faced:

Missing auth.log on Kali.

GPG key errors during rsyslog installation.

Resolved by updating sources, installing kali-archive-keyring, and reinstalling rsyslog.



Key Takeaways:

1. auth.log is critical for detecting SSH attacks.
2. Persistence and troubleshooting are essential when configuring logging on Linux.

Final Reflections

1. Log analysis is essential for detecting attacks, investigating incidents, and building strong defenses.
2. Mastering Windows Event IDs, Linux logs, and basic detection patterns empowers SOC analysts to stay ahead of threats.